Riddhiman K. Garge, Ph.D.

Personal Contact Details

Email	rgarge@uw.edu, riddhiman92@gmail.com
Phone	+1-737-529-2534
Bluesky	@riddhiman.bsky.social
Google Scholar	https://scholar.google.com/citations?user=ZDxv9l8AAAAJ&hl=en
ORCID	https://orcid.org/0000-0002-6774-0172

Education

- 2015 - 2021 **Ph.D., Biochemistry**
  The University of Texas (UT) at Austin
  Advisor: Dr. Edward M. Marcotte
  Thesis entitled "Of Yeast and Men : Insights into Evolution and Human Health from 1 Billion Years of Divergence"

- 2011 - 2015 **B.E., Biotechnology**
  M. S. Ramaiah Institute of Technology, Bangalore

Positions and Research Experience

- 2021 - Present **Postdoctoral Researcher**
  Center for Multiplex Assessment of Phenotype (CMAP), Department of Genome Sciences, University of Washington
  Advisors- Drs. Jay A. Shendure & Lea M. Starita

- 2021 - Present **Research Fellow**
  Brotman Baty Institute for Precision Medicine, University of Washington
  Advisors- Drs. Jay A. Shendure & Lea M. Starita

- 2016 - 2021 **Ph.D. Student**
  Department of Molecular Biosciences, The University of Texas at Austin
  Advisor- Dr. Edward M. Marcotte

- 2015 - 2016 **Rotation Student**
  Department of Molecular Biosciences, The University of Texas at Austin
  Advisors- Drs. Edward M. Marcotte, Andreas Matouschek, Rick Russell

- 2014 - 2015 **Research Assistant**
  Institute of Stem Cell and Regenerative Medicine, National Center for Biological Sciences, Bengaluru
  Advisors- Dr. Ramaswamy Subramanian & Dr. Jeffrey Abramson

- 2013 **Research Assistant**
  Bugworks Research India Pvt Ltd.
  Advisor- Dr. Bala Subramanian

- 2012 **Research Assistant**
  Schrödinger India Inc.
  Advisor- Dr. Ravikumar Muttineini

Fellowships and Awards

- 2023 - Present Washington Research Foundation Postdoctoral Fellowship
- 2021 - 2023 CMAP Postdoctoral Fellowship

- 2019 Institute for Cellular and Molecular Biology (ICMB) Travel Award
- 2018 - 2020 American Heart Association (AHA) Predoctoral Fellowship
- 2018 Marion Elizabeth Eason Endowed Scholarship for The Study of Biology
- 2018 Graduate Dean's Prestigious Fellowship Supplement
- 2018 David Bruton Jr. Graduate School Fellowship
- 2018 ICMB Biochemistry Program Summer Travel Award
- 2017 ICMB Biochemistry Program Summer Travel Award

PUBLICATIONS

** denotes equal contribution, ^ denotes co-corresponding author*

1. Fayer, S., **Garge, R.K.**, Hopkins, M., Friedman, C.E., McGee, A.V., Rico, J., Powell, R. L., McDermot, E., Smith N.T., Pendyala, S., Richardson, M.E., Smith, E.D., et al. Editing stem cell genomes at scale to measure variant effects in diverse cell and genetic contexts. ***medRxiv*** (2025)

2. Woo, I., Casadei, S., Snyder, M.W., Smith, N.T., Best, S., Tejura, M., Gupta, P., McEwen, A.E., Post, M., Hamm, A., Dawood, M., Hosokai A., Alicia X., **Garge R.K.**, et al. Saturation genome editing of BARD1 resolves VUS and provides insight into BRCA1-BARD1 tumor suppression. ***medRxiv*** (2025).

3. Pinglay, S., Lalanne, J.-B., Daza, R.M., Kottapalli, S., Quaisar, F., Koeppel, J., **Garge, R.K.**, Li, X., Lee, D.S., and Shendure, J.Multiplex generation and single-cell analysis of structural variants in mammalian genomes. ***Science*** (2025)***.***

4. **Garge, R.K.**^, Lynch, V., Fields, R., Casadei, S., Best, S., Stone, J., Snyder, M., McGann, C.D., Shendure, J., Starita, L.M., et al. The proteomic landscape and temporal dynamics of mammalian gastruloid development. ***bioRxiv, In revision*** (2024).

5. Hamazaki, N., Yang, W., Kubo, C., Qiu, C., Martin, B.K., **Garge, R.K.**, Regalado, S.G., Nichols, E., Lee, C., Daza, R.M., et al. Induction and *in silico* staging of human gastruloids with neural tube, segmented somites & advanced cell types. ***Nature Cell Biology*** (2024).

6. Boyle, G.E., Sitko, K., Galloway, J.G., Haddox, H.K., Bianchi, A.H., Dixon, A., Thomson, R.E.S., **Garge, R.K.**, Rettie, A.E., Rubin, A.F., et al. Deep mutational scanning of CYP2C19 reveals a substrate specificity-abundance tradeoff. ***Genetics*** (2024).

7. **Garge, R.K.**, Geck, R.C., Armstrong, J.O., Dunn, B., Boutz, D.R., Battenhouse, A., Leutert, M., Dang, V., Jiang, P., Kwiatkowski, D., et al. Systematic Profiling of Ale Yeast Protein Dynamics across Fermentation and Repitching. ***G3: Genes, Genomes, Genetics*** (2023).

8. Abdullah, M., Greco, B.M., Laurent, J.M., **Garge, R.K.**, Boutz, D.R.,Vandeloo, M., Marcotte, E.M., Kachroo, A., Rapid, Scalable, and Combinatorial Genome Engineering by Markerless Enrichment and Recombination of Genetically Engineered Loci (MERGE) in Yeast. ***Cell Reports Methods*** (2022).

9. Bean, B.D.M., Mulvihill, C.J., **Garge, R.K.**, Boutz, D.R., Rousseau, O., Floyd, B.M., Cheney, W., Gardner, E.C., Ellington, A.D., Marcotte, E.M., et al. Functional expression of opioid receptors and other human GPCRs in yeast engineered to produce human sterols. ***Nature Communications*** (2021).

10. Roberson E., Battenhouse A., **Garge R. K.**, Tran N. K., Marcotte E. M., Wallingford J. B., Spatiotemporal transcriptomics of the cycling mouse oviduct. Cover article. ***Developmental Biology*** (2021).

11. **Garge, R.K.***, Cha, H.J.*, Lee, C., Gollihar, J.D., Kachroo, A.H., Wallingford, J.B., and Marcotte, E.M. Antifungal benzimidazoles disrupt vasculature by targeting one of nine β-tubulins. Cover article. ***Genetics*** (2020).

12. **Garge, R.K.**, Laurent, J.M., Kachroo, A.H., and Marcotte, E.M. Systematic Humanization of the Yeast Cytoskeleton Discerns Functionally Replaceable from Divergent Human Genes. Cover article. ***Genetics****, 1153–1169* (2020).

13. Laurent, J.M., **Garge, R.K.**, Teufel, A.I., Wilke, C.O., Kachroo, A.H., and Marcotte, E.M. Humanization of yeast genes with multiple human orthologs reveals functional divergence between paralogs. ***PLoS Biology*** (2020).

14. Blank, H.M., Papoulas, O., Maitra, N., **Garge, R.K.**, Kennedy, B.K., Schilling, B., Marcotte, E.M., and Polymenis, M. Abundances of transcripts, proteins, and metabolites in the cell cycle of budding yeast reveal coordinate control of lipid metabolism. ***Molecular Biology of the Cell*** (2020).
15. Akhmetov, A., Laurent, J.M., Gollihar, J., Gardner, E.C., **Garge, R.K.**, Ellington, A.D., Kachroo, A.H., and Marcotte, E.M. Single-step Precision Genome Editing in Yeast Using CRISPR-Cas9. ***Bio Protocol*** (2018).

Press

- 2024 Promising Young Scientist: Riddhiman Garge Embraces Two Distinct Passions
- 2024 Annual Intern Forum-Luncheon Provides Students Opportunities to Discuss Careers with Scientists
- 2024 Brü Lab Podcast Ep.142- Changes in Yeast Proteins During Fermentation with Dr. Riddhiman Garge
- 2024 Beer brewers and geneticists collaborate on study of yeast changes during commercial fermentation
- 2021 How an anti-fungal medication can stop new blood vessel formation

Conferences and Presentations

2025	Invited Seminar	UW Genome Sciences Department
2024	Invited Seminar	Biotechnology Department, Ramaiah Institute of Technology
2024	Poster	NHGRI Centers in Genome Excellence (CEGS) Meeting
2024	Organizer	Alpine Genomics Institute, Challenges and Opportunities in Genetics and Genomics
2024	Invited Seminar	Dell Pediatric Research Institute, UT Austin
2024	Invited Seminar	Biochemistry and Molecular Genetics Department, CU Anschutz
2024	Lightning Talk	US Human Proteome Organization
2023	Invited Talk	UW Institute of Stem Cell and Regenerative Medicine seminar
2023	Participant	NHGRI Centers in Genome Excellence (CEGS) Meeting
2023	Invited Talk	Master Brewers Association of the Americas (MBAA) Conference
2023	Poster	Advances in Genome Biology and Technology (AGBT) Precision Health Meeting
2023	Poster	82nd Annual Society for Developmental Biology Meeting
2022	Poster	NHGRI Centers in Genome Excellence (CEGS) Meeting
2020	Participant	From Functional Genomics to Systems Biology, EMBL
2020	Poster	Molecular Mechanisms in Evolution and Ecology, EMBL
2020	Poster	The Allied Genetics Conference
2019	Poster	Yeast Research: Origins, Insights & Breakthroughs, CSHL
2019	Poster	American Society of Brewing Chemists Conference
2018	Poster	From Functional Genomics to Systems Biology, EMBL
2018	Poster	Genetics Society of America Yeast Genetics Meeting
2017	Participant	DNA23
2017	Selected Talk	Society for Molecular Biology and Evolution
2017	Participant	5th Annual Big Data in Biology Symposium at UT Austin
2016	Participant	4th Annual Big Data in Biology Symposium at UT Austin
2014	Poster	Recent Advances in Computational Drug Design, Indian Institute of Science

Teaching Experience

2020	Teaching Assistant	Introduction to Core Next Generation Sequencing Concepts and Tools
2018	Teaching Assistant	Systems Biology and Bioinformatics

WORKSHOPS

- 2019 Principles of Machine Learning for Bioinformatics
- 2018 Introduction to Core Next Generation Sequencing Concepts and Tools
- 2016 Introduction to Mass Spectrometry and Proteomics

MENTORING

( ) denotes current position

- **Undergraduate students**

1.	2024	Clare Malhotra	REU student (Undergraduate at University of Chicago)
2.	2022-2023	Annabelle Fryer	Research Assistant (Patient Services Coordinator II Massachusetts General Hospital)
3.	2021-Present	Ivan Woo	Research Assistant
4.	2021-2025	Audrey Hamm	Research Assistant
5.	2017-2019	Sophie Curie	Research Assistant (UW Radiology Resident)
6.	2018	Anastasiya Kulikova	Research Assistant (UT Austin Molecular Biosciences graduate student)

- **Graduate students**

1.	2025	Melissa Hopkins	Rotation student (UW Genome Sciences graduate student)
2.	2025	Alexandra Donny	Rotation student (UW Genome Sciences graduate student)
3.	2024	Abby McGee	Rotation student (UW Genome Sciences graduate student)
4.	2023	Evan McDermott	Rotation student (UW Genome Sciences graduate student)
5.	2022	Connor Kubo	Rotation student (UW Genome Sciences graduate student)
6.	2017	Evelyn Abbott	Rotation student (UT Austin Ecology and Evolutionary Biology graduate student)

SERVICE

• 2024	Judge and Guest of Honor	Sadhu Vasvani International School Science Fair
• 2024	Invited Panelist	Brotman Baty Institute Career Panel for Undergraduates
• 2024	Organizing committee	Alpine Genomics Institute Unconference
• 2024	Graduate Recruitment Panel	UW Department of Genome Sciences
• 2022	Organizing Member	UW Department of Genome Sciences 20th Anniversary Symposium
• 2019	Committee Member	Gottlieb Lecture Series featuring Dr. Jay Shendure
• 2017-2020	Organizer	"Biology Talks about Computers, Organisms and Systems", Center for Systems and Synthetic Biology, UT Austin
• 2016	Committee Member	Gottlieb Lecture Series featuring Dr. William Moerner
• 2016	Judge	Molecular Biosciences Undergraduate Research Symposium, UT Austin

SCIENTIFIC OUTREACH

• 2025	Volunteer	UW Robinson Center's visit to the Brotman Baty Institute
• 2023	Invited guest	The Brü Lab podcast hosted by Brülosophy
• 2018	Volunteer	Magellan International School visit to UT Austin
• 2017	Invited guest	NPR's "They Blinded Me With Science" on KVRX
• 2016	Volunteer	NSF's Building with Biology event at the Thinkery in Austin